

EPVS — Egg Production Verification System

Process Flow, User Roles & Business Logic

1. User Roles & Permissions

The system uses five roles, each with distinct access levels and responsibilities.

1.1 Super Admin

- Full system access — can manage all EPVs, users, clients, invoices, and reconciliation.
- Can send EPV forms to clients via email.
- Can approve/reject EPVs, reconcile payments, and manage KPI targets.
- Access to Dashboard, Inspector Dashboard, and Administrator Dashboard.
- Can assign inspectors to provinces and manage all user accounts.

1.2 Admin

- Similar access to Super Admin for day-to-day operations.
- Can manage EPVs, reconciliation, and view all dashboards.
- Cannot modify KPI targets (Super Admin only).

1.3 Inspector

- Assigned to a specific province — can only see facilities in that province.
- Primary role: verify (approve or reject) completed client EPVs.
- When rejecting, creates an "Inspector EPV" with the correct figures.
- Can mark facilities as physically visited (Manual Inspection).
- Access limited to the Inspector Dashboard.

1.4 Company Admin

- Company-level user linked to specific facilities.
- Can view their Company Overview with allocated facilities and EPVs.
- Can create manual EPVs for past months, complete EPV forms, and upload Proof of Payment.
- Cannot access Inspector or Administrator dashboards.

1.5 User (Company User)

- Basic company-level access — read and submit EPVs.
- Can complete EPV forms sent to them and upload Proof of Payment.
- Cannot access Inspector or Administrator dashboards.

2. EPV Process Flow (End-to-End)

2.1 EPV Creation

An EPV can be created in two ways:

- **A) Sent by Super Admin / Admin:** From the Client Allocation page, click "Send EPV" for a facility. The system generates a unique reference number (EPV-YYYY-MM-XXXX), creates a Pending record, and emails a secure link to the facility's contacts.
- **B) Manual Creation by Company Admin:** From the Company Overview, a Company Admin can create an EPV for any past month that doesn't already have one.

2.2 Business Rules for Creation

- Only one EPV per facility per month — duplicates are blocked.
- Future months are not allowed for manual creation.
- Opening stock is auto-populated from the previous month's closing stock.
- A unique reference number is generated: EPV-YYYY-MM-XXXX (sequential).

2.3 Form Completion (4 Steps)

The client (or company user) fills out the EPV form via a secure token link or from the Company Overview:

- **Step 1 — Business Details:** Business name, facility type, province, authorized person, contact numbers, and email.
- **Step 2 — Egg Levy:** Opening stock, purchases (graded/ungraded), deductions (market returns, machine loss, sent to pulp, destroyed), and sales (trade, staff, farm stall). System calculates closing stock and levy.
- **Step 3 — Pulp Levy:** Pulp opening stock, purchased, converted from eggs, sold to trade, sold to producers. System calculates pulp levy.
- **Step 4 — Review & Submit:** Review all entries and submit. Status changes from Pending to Completed.

2.4 Levy Calculation

- **Egg Levy:** $\text{Total Sales (SoldToTrade)} \times 0.018$ (1.8%)
- **Pulp Levy:** $\text{PulpSoldToTrade} \times 1.7$ (conversion to dozens) $\times 0.018$
- **Total Levy:** Egg Levy + Pulp Levy — this is the amount owed by the facility.

3. Inspector Dashboard — What Shows Up & Why

The Inspector Dashboard shows EPV data filtered by the inspector's assigned province (or all provinces for Super Admin/Admin).

3.1 Pending Approvals

These are EPVs waiting for the inspector to verify. An EPV appears here when ALL of the following are true:

- EPVType = "Client" (submitted by the facility, not an inspector EPV).
- Status = "Completed" (the client has filled out and submitted the form).
- IsVerified = 0 or NULL (not yet approved by an inspector).
- No Inspector EPV has been created for this EPV yet (no rejection record exists).
- The facility's province matches the inspector's assigned province.

Additionally, EPVs where an Inspector EPV exists but the Inspector EPV's Status is still "Pending" also appear — these were previously rejected and the inspector hasn't completed their own form yet.

3.2 Inspector Actions

- **Approve:** Sets IsVerified = 1, records VerifiedBy and VerifiedAt. The EPV then becomes eligible for reconciliation on the Admin side.
- **Reject:** Creates a new EPV record with EPVType = "Inspector", linked to the original via LinkedEPVId. The inspector then fills out their own version of the form with the correct production figures.

3.3 Inspector EPVs to Complete

After rejecting a client EPV, the inspector must complete their own form. These appear when:

- EPVType = "Inspector" (created as a rejection counterpart).
- Status = "Pending" (inspector hasn't completed their form yet).
- InspectorId matches the logged-in inspector.

3.4 Facilities Needing Visit This Quarter

Shows facilities where no physical inspection (ManualInspection = 1) has been recorded in the current quarter.

- Facilities in the inspector's assigned province.
- No EPV exists for this quarter with ManualInspection = 1.

3.5 Not Completed EPVs

Shows facilities that have NOT submitted a completed EPV for each month of the current year:

- For each month from January to the current month, facilities without a Completed Client EPV are listed.

3.6 Facilities Owning Money

Shows facilities with outstanding balances:

- EPVType = "Client" and Status = "Completed".
- Total Billed (Egg Levy + Pulp Levy) minus Reconciled Amount > 0.
- Grouped by facility, showing total outstanding.

3.7 Inspector KPIs

KPI	Formula	Default Target
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Collection Rate	$(\text{Total Paid} / \text{Total Billed}) \times 100$	80%
Approvals Actioned	$(\text{Verified Count} / \text{Total EPVs}) \times 100$	90%
Facilities Visited	$(\text{Visited} / \text{Total Facilities}) \times 100$	100%

4. Administrator Dashboard — What Shows Up & Why

The Administrator Dashboard focuses on financial reconciliation of completed, verified EPVs.

4.1 Reconciliation Data

EPVs appear on the Administrator Dashboard when:

- EPVType = "Client" (only client-submitted EPVs, not inspector copies).
- Status = "Completed" (the form has been submitted).

EPVs are categorized into three tabs:

- **Outstanding:** IsReconciled = 0 (or NULL) AND the outstanding amount (Total Billed – Reconciled Amount) > 0. These need payment.
- **Partial:** IsReconciled = 0 AND ReconciledAmount > 0 but less than Total Billed. Some payment has been received but not the full amount.
- **Reconciled:** IsReconciled = 1. The full amount has been received and marked as reconciled.

4.2 Reconciliation Process

The administrator can reconcile EPVs individually or in batch:

- **Single Reconciliation:** Enter the amount received for one EPV. If the amount $\geq$ Total Billed, the EPV is marked fully reconciled (IsReconciled = 1). Otherwise it stays as partially reconciled.
- **Batch Reconciliation:** Select multiple EPVs, enter amounts for each (or click "Full Amount"), and reconcile all at once.
- **Mark Full Payment:** One-click to set ReconciledAmount = Total Billed and IsReconciled = 1.

4.3 Proof of Payment (POP)

The payment verification process:

- Client uploads POP (PDF/PNG/JPG, max 10MB) via Company Overview.
- On upload, ClientPaymentStatus is automatically set to "Paid".
- If POP is deleted, ClientPaymentStatus resets to "Not Paid".
- Admin can view the uploaded POP file and add comments.
- POP upload is NOT required for admin to reconcile — Super Admin/Admin can reconcile regardless.

4.4 Administrator KPIs

KPI	Formula	Default Target
Collection Rate	$(\text{Total Paid} / \text{Total Billed}) \times 100$	80%
Reconciliation Rate	$(\text{Reconciled Count} / \text{Total EPVs}) \times 100$	90%

Outstanding Rate	$(\text{Outstanding} / \text{Total Billed}) \times 100$ — lower is better	$\leq 5\%$
Verification Rate	$(\text{Verified Count} / \text{Total EPVs}) \times 100$	90%

Note: KPI targets can be edited by Super Admin from the main Dashboard.

5. EPV Status Flow Summary

Stage	Status / Flag	Triggered By
1. EPV Created	Status = "Pending"	Super Admin sends EPV or Company Admin creates manually
2. Form Submitted	Status = "Completed"	Client/Company User fills out and submits the 4-step form
3a. Inspector Approves	IsVerified = 1	Inspector clicks "Approve" — EPV is now eligible for reconciliation
3b. Inspector Rejects	Inspector EPV created (Status = "Pending")	Inspector clicks "Reject" — must complete their own EPV form
4. POP Uploaded	ClientPaymentStatus = "Paid"	Client uploads Proof of Payment file
5a. Partial Reconciliation	ReconciledAmount > 0, IsReconciled = 0	Admin enters partial payment amount
5b. Full Reconciliation	IsReconciled = 1	Admin enters full payment amount or clicks "Mark Full"

6. Reference Numbers & Audit Trail

6.1 Reference Number Format

- **Format:** EPV-YYYY-MM-XXXX
- YYYY = Year, MM = Month (zero-padded), XXXX = Sequential counter (0001–9999).
- Example: EPV-2026-03-0042 = the 42nd EPV created for March 2026.

6.2 Audit Trail

All changes are tracked in the EPVAuditLog and ClientAuditLog tables:

- Every field change records: which EPV, which field, old value, new value, who changed it, and when.
- Tracked actions include: EPV sends, form submissions, approvals, rejections, POP uploads, reconciliation, and manual inspections.

7. Province Tracking

- Each facility has a FacilityProvince assigned in the master database.
- Each Inspector is assigned an InspectorProvince — they can only view/manage facilities in their province.

- Super Admin and Admin can filter by province or view all provinces.
- All statistics, charts, and KPIs can be filtered by province.
- Nine provinces: Eastern Cape, Free State, Gauteng, KwaZulu-Natal, Limpopo, Mpumalanga, North West, Northern Cape, Western Cape.

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Recent Updates - April 2026

This section summarises the changes shipped in April 2026. Earlier sections of this manual remain accurate; the items below override or extend them.

EPV period now reflects the previous month

Whenever an EPV is issued (manually from the Consolidated Master Facility Database or Company Overview, or automatically each month), the verification is created for the previous calendar month.

Example: an EPV issued in March 2026 captures February 2026 data.

Automatic monthly EPV sending

Once a facility has been placed on the EPV cycle (status 'On EPV Cycle'), the system now automatically issues that facility a fresh EPV at the start of every month covering the previous month's data. The scheduler runs once a day; if the previous-month EPV is already present, nothing is sent. The email goes to all four facility addresses on file (primary, owner, accounts and manager). To take a facility off the cycle, clear its EPVCycleStatus on the Consolidated Master Facility Database page.

Resend EPV email button

The Company Overview window now shows a 'Resend Email' button alongside every Pending EPV. Use it when a facility reports they never received the original notification - it re-sends the same EPV link to all facility email addresses on file. No new EPV record is created and the resend is logged in EmailSendLog.

Facility verified status now visible

The Consolidated Master Facility Database page shows a 'Verified' badge once the facility's invited Company Admin has accepted their invitation and completed the onboarding wizard. A separate 'On EPV Cycle' tag indicates the facility is on the monthly auto-send schedule. The status cell stays empty until one of those two events applies.

EPV form (Eggs page) - new 'Eggs Exported' deduction

Section C (Deductions) now includes 'Eggs Exported'. Like the other deductions, it reduces the theoretical closing stock and does NOT affect the egg levy (only Section D Sales drives the levy).

EPV form (Eggs page) - purchase source detail and attachments

When Graded Eggs Purchased or Ungraded Eggs Purchased is greater than zero, a 'Source & Supplier Detail' comment box opens under Section B together with a file uploader. Facility users and inspectors can describe where and from whom the purchase was made, and attach supporting documents (PDF, PNG, or JPG, up to 15 MB each). Uploaded files appear with a 'Remove' button while the EPV is still editable, and as inline links in the Review section.

EPV form (Pulp page) - 'Conversion Loss' line

Sales & Deductions on the pulp page now has a 'Conversion Loss' row directly below 'Sold to Other Producers'. Conversion loss does NOT contribute to the pulp levy, but it is subtracted from closing pulp stock so the carry-forward stays balanced.

EPV form (Pulp page) - pulp purchase source detail and attachments

When 'Pulp Purchased from Others' is greater than zero, a comment box and uploader appear (same 15 MB limit and PDF/PNG/JPG rules) so the source of the purchased pulp can be recorded and supporting documents attached.

Verifications can only be captured for completed months

The 'Add Verification' modal in Company Overview now defaults to the previous month, and the month picker hides the current and any future months. The backend rejects current-month submissions with a clear error message: a month must be over before its EPV can be captured.